EMULSIFIER

Film Emulator & Rendering Engine

Official User Manual & Color Science Guide

Version 1.79

Introduction

Welcome to Emulsifier. Digital sensors and AI image generators capture light linearly and perfectly, creating a sterile, "plastic" aesthetic. Physical film, however, is a messy, subtractive medium. It relies on imperfect chemical dye layers, microscopic silver halide crystals, and optical glass that scatters light.

Emulsifier does not use cheap "video game filters" or simple digital LUTs. Instead, it pushes your image through a simulated physical pipeline, converting your linear digital exposure into the logarithmic density domain of real film. This manual will guide you through the interface and the physics behind each tool, helping you craft authentically cinematic imagery.

Part 1: General Interface & Workflow

LOADING IMAGES & EXPORTING

> **Loading Digital / AI Photos:** Click the prominent button at the top right, or simply drag and drop any JPG, PNG, or TIFF file directly into the application window. Images larger than 1024px are automatically proxied down for real-time UI performance.
>
> **Exporting Full-Res Renders:** The "Export Full-Res Render" button at the bottom right bypasses the proxy. It takes your original, massive image file and funnels it through the exact mathematical coordinates of your sliders, saving a flawless, uncompressed TIFF, PNG, or high-quality JPG.

CHOOSING THE TONAL CURVE

> **The Dropdown Menu:** This menu scans your local `/profiles` folder for JSON/CSV densitometry data. Selecting a profile mathematically maps your digital image's luminance into the exact physical density curve of a real-world film stock (like Kodak Vision3 or Fuji Provia).

THE DUAL-LUMINANCE HISTOGRAMS

> **Ghost & Gold:** The scope at the top right is your objective anchor. The **Grey "Ghost"** shape represents the luminance structure of your original digital image. It never moves. The **Gold** shape represents your dehanced image in real-time. This allows you to visually monitor if your film curve is crushing shadows or blowing out highlights compared to the original.

COMPARATIVE VIEW MODES

> At the bottom right of the image preview, you can toggle how you compare your edit to the original:
>
> - **Side-by-Side [|]:** Splits the screen evenly. Left is original, right is dehanced. Best for overall color-grading judgment.
> - **Swipe (WIPE):** Overlays the dehanced image over the original. Click and drag the vertical divider. Crucial for comparing precise pixel details, like skin tones or halation bleed, in the exact same geographic spot.

ZOOM MODES & PANNING

> **Zoom:** Use your mouse's **scroll wheel** to smoothly punch in and out of the image for detailed inspection. Alternatively, click anywhere on the canvas to instantly punch into a 1:1 pixel view (mandatory for accurately judging fine details like Crystal Size and Chromatic Aberration). Click again to zoom out to "FIT".
>
> **Panning:** While zoomed in, hold the **Spacebar**. Your cursor will turn into a hand, allowing you to click and drag to pan around the image.

UNDO / REDO (NON-DESTRUCTIVE EDITING)

> Emulsifier remembers your last 5 parameter changes. Use standard OS shortcuts (**Ctrl+Z / Cmd+Z** to Undo, and **Ctrl+Y / Cmd+Y / Cmd+Shift+Z** to Redo). When triggered, a gentle, semi-transparent "glass" OSD (On-Screen Display) will fade onto the center of the image to confirm the state change, preventing you from getting lost in your edit history.

OVERALL PHYSICAL MIX & SMART AUTO-MIX

> **The Concept:** This is the master opacity of your film grade. Because film density is highly aggressive, a 100% mix often washes out digital midtones.
>
> **The Auto Button:** Clicking "Auto" triggers the Global Washout Detector. The engine silently tests multiple mix strengths in the background, analyzing the overall brightness of the image. It automatically stops at the absolute highest percentage that protects your midtones from washing out by more than 6%. *Always start your grade by clicking Auto.*

Part 2: The Physics Pipeline

The sliders in Emulsifier are executed in a strict chronological pipeline, mimicking the physical journey of light hitting a lens, exposing the emulsion, and being printed in a darkroom.

1. LENS OPTICS

Simulates the physical glass of vintage taking lenses.

Optical Softness

- **Analog Parallel:** Vintage lenses lack the hyper-sharp micro-contrast and resolving power of modern digital sensors.
- **Crucial Because:** Digital sharpness feels clinical and artificial. Blurring the image mathematically binds the pixels together before the grain is applied.
- **What it Alters:** Applies a targeted Gaussian blur calculated against the master image width.
- **Examples:** At **0%**, the image remains digitally sharp. At **100%**, it resembles a heavily out-of-focus dream sequence. *Advisory: Use between 10-20% to make digital skin look creamy and organic.*

Chromatic Aberration

- **Analog Parallel:** Lateral color fringing caused by glass dispersion, where different wavelengths of light hit the lens at different speeds, splitting red and blue channels near high-contrast edges.
- **Crucial Because:** Imperfect glass is a hallmark of anamorphic and vintage spherical cinema lenses.
- **What it Alters:** Physically scales and shifts the Red and Blue channels in opposite directions from the image center.
- **Examples:** At **0px**, colors are perfectly aligned. At **10px**, heavy 3D-glasses-style splitting occurs. *Advisory: Keep under 3px for a subtle, subconscious cinematic feel.*

2. LIGHT & SCATTER

Simulates how light bounces around inside the lens barrel and the physical film emulsion.

Cine-Log Flattening

- **Analog Parallel:** Film scans traditionally arrive in a flat, "Log" color space to preserve dynamic range before grading.
- **Crucial Because:** If you feed a highly contrasted, punchy digital JPG into a high-contrast film profile, the shadows will instantly crush into black voids.
- **What it Alters:** Applies a logarithmic curve to lift shadows and compress highlights prior to the film node.
- **Examples:** At **0%**, digital contrast remains intact. At **100%**, the image becomes highly washed out and milky. *Advisory: If your film profile makes the image too dark, push this to 25-40% to feed "softer" light into the emulsion.*

Emulsion Crosstalk

- **Analog Parallel:** In real film, the Cyan, Magenta, and Yellow dye layers are chemically impure. Exposing the green layer often accidentally "bleeds" into the red layer.
- **Crucial Because:** This creates complex, subtractive hue twists that are mathematically impossible to replicate with standard digital RGB hue sliders.
- **What it Alters:** Pushes the image through a 3x3 color matrix, blending opposing channels.
- **Examples:** At **0%**, colors are true to life. At **100%**, greens violently shift to deep cyan, and reds push to thick orange. *Advisory: 20-30% instantly yields a rich, vintage "Kodak" color palette.*

Dual-Stage Halation

- **Analog Parallel:** Bright light travels through the emulsion, bounces off the shiny backplate of the camera, and scatters back into the bottom-most Red dye layer of the film.
- **Crucial Because:** Halation is the most recognizable signature of analog motion picture film, providing a magical "glow" to light sources.
- **What it Alters:** Isolates the top 60% of highlights, blurs them using a narrow (core) and wide (falloff) radius, tints them crimson, and adds them back to the image.
- **Examples:** At **0%**, lights are stark white. At **100%**, highlights bleed intense red/orange into the surrounding shadows. *Advisory: Highly effective on neon signs, practical lamps, and high-contrast sky edges.*

Lens Bloom

- **Analog Parallel:** Light scattering across the glass elements inside the lens barrel itself, often enhanced by using a "Pro-Mist" or diffusion filter.

- **Crucial Because:** It blooms the highlights softly without the harsh red tint of halation, lowering global micro-contrast.
- **What it Alters:** Generates a very wide, neutral-luminance blur derived from the highlight thresholds.
- **Examples:** *Advisory: Combine 15% Bloom with 20% Halation for a glowing, dream-like cinematic atmosphere.*

3. DARKROOM PRINT

Controls how the "developed negative" is printed onto photographic paper.

Analog Print Contrast

- **Analog Parallel:** Simulates choosing different numerical grades of photographic printing paper in a darkroom (Grade 1 is flat, Grade 5 is harsh).
- **Crucial Because:** Standard digital contrast increases saturation unnaturally. Analog contrast steepens the image density organically without making colors neon.
- **What it Alters:** Applies a localized S-Curve formula specifically tailored to film density mapping.
- **Examples:** At **0%**, the base film profile is untouched. At **100%**, blacks are aggressively crushed and highlights pop. *Advisory: Use to bring "snap" and depth back into a flat image.*

Subtractive Saturation

- **Analog Parallel:** In physical film, pure white (burnt paper) and pure black (thick dye) physically cannot hold color.
- **Crucial Because:** Digital algorithms allow bright skies to be 100% saturated neon blue. Film natively desaturates highlights and shadows, rolling them off into pleasing pastels.
- **What it Alters:** Applies a parabolic luminance mask that geometrically reduces color saturation at the extreme ends of the histogram.
- **Examples:** At **0%**, digital neon colors remain. At **100%**, bright skies and dark shadows turn almost grayscale, pushing color exclusively into the midtones.

Warm/Cool Split Tone

- **Analog Parallel:** Slight chemical temperature imbalances during darkroom development.
- **Crucial Because:** Adds mood and color-separation to the grade without touching white-balance.
- **What it Alters:** A photometric push that maps warm vectors (yellow/red) to highlights and cool vectors (cyan/blue) to shadows, or vice-versa depending on slider direction.
- **Examples:** Pushing positive (**+25%**) creates a nostalgic, golden-hour warmth. Pushing negative (**-25%**) creates a moody, modern "teal-and-orange" cinematic push.

4. PHYSICAL GRAIN

Simulates the physical silver halide crystals that make up the image.

Dye Cloud Amount & Crystal Size

- **Analog Parallel:** Film isn't made of perfectly square pixels; it is comprised of randomly scattered, irregular clumps of silver halide.
- **Crucial Because:** True film grain is non-linear—it is highly visible in midtones, but disappears in pure blacks and pure whites. Digital "noise" overlays ignore this physics and ruin shadows.
- **What it Alters:** Generates algorithmic noise, scales it based on the virtual width of the master image, and masks it aggressively through a midtone luminance curve.
- **Examples:** For a clean, modern 35mm motion picture look, use **Amount 25%** and **Size 1.2**. For a gritty, high-ISO 16mm indie-film aesthetic, push to **Amount 60%** and **Size 2.8**.

5. EDGE IMPERFECTIONS

Field Flatness Softness & Creep

- **Analog Parallel:** Older, cheaper lenses lose resolving power and become blurry toward the outer edges of the glass.
- **Crucial Because:** It subconsciously draws the viewer's eye to the sharp center subject.
- **What it Alters:** Softness controls the blur intensity; Creep controls how far the radial mask pushes into the center of the frame.

Vignette Intensity & Creep

- **Analog Parallel:** Physical light falloff caused by the lens barrel blocking light from hitting the corners of the film plane.
- **What it Alters:** A smooth radial exposure reduction. Use **Intensity 30%** and **Creep 40%** to naturally and subtly frame a subject.

6. OUTPUT LEVELS

The Concept: This is a final, purely digital adjustment pass occurring at the very end of the pipeline. Use the three interactive triangles beneath the histogram to compress or stretch the tonal range of the final output. Right-click anywhere on the tool to reset it.

- **Black Point (Left Triangle):** Sets the absolute black level. Authentic film emulations often leave the shadows slightly raised ("milky"). Slide this slightly to the right to anchor true digital black (RGB 0,0,0) into your image without destroying the filmic midtones.
- **Midtone / Gamma (Center Triangle):** Adjusts the global brightness of the middle gray values. For example, if the film emulation leaves the mids too bright or washed out, slide this to the right to compress them down to darker, richer levels.
- **White Point (Right Triangle):** Sets the absolute peak highlight. If the film emulation severely dulls your highlights, slide this left to stretch the brightest pixels back up to pure digital white.